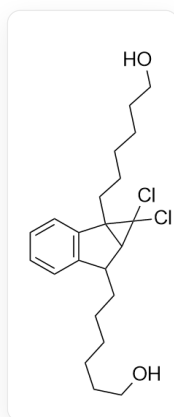


Question

Under external mechanical forces, some polymers can undergo chemical reactions due to their special groups; such polymers are called mechano-responsive polymers.

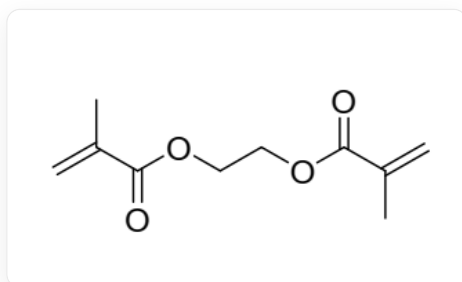
There are three compounds **1**, **2**, **3**, **4**, and their structures are as follows:

1:



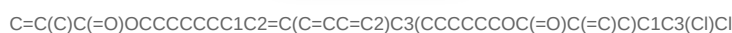
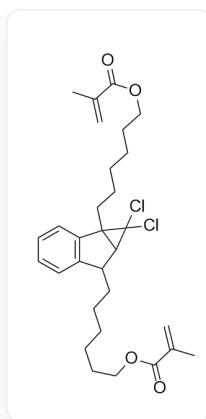
C(CCCO)CCC1C2=C(C=CC=C2)C3(CCCCCO)C1C3(Cl)Cl

2:

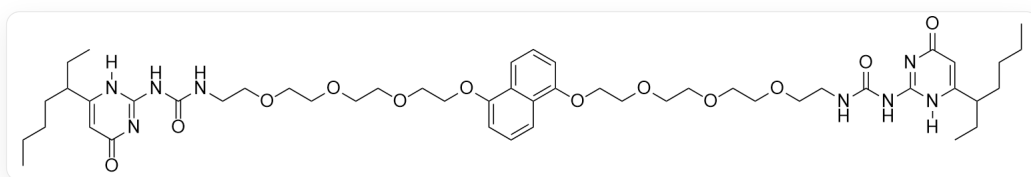


C=C(C)C(=O)OCCOC(=O)C(=C)C

3:



4:



Known: Heating **1** at 50°C for 24h will release HCl to obtain an oily liquid; methyl acrylate is polymerized with benzoyl peroxide as an initiator. If only a small amount of **1** (~ 5%) and **2** (~ 5%) are mixed in the system, polymer **A** is obtained; if only a small amount of **3** (~ 5%) is mixed in the system, polymer **B** is obtained. Experiments show that **A**, **B** can release HCl when pressurized, but the amount is different; **4** exists as a cyclic monomer in the solid state, and when it is dissolved in chloroform, when the concentration exceeds a certain value, the viscosity of the solution increases sharply with the concentration. Please select the correct option from the following.

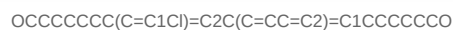
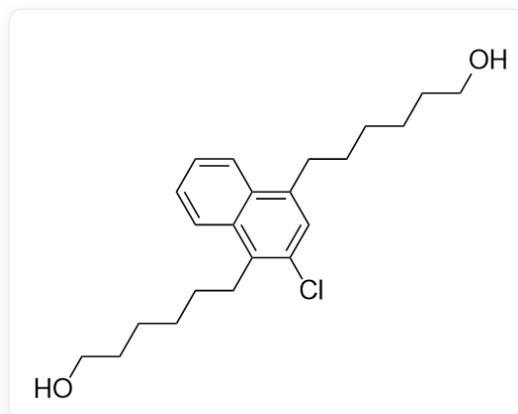
- A.** The stoichiometric ratio of HCl produced by the complete reaction of heating **1** to the resulting product molecules is 2:1.
- B.** The aromatic system present in the product molecule obtained by heating **1** contains 6 π electrons.
- C.** **A** has a covalently cross-linked network structure while **B** does not.
- D.** Under the same pressure, an equal amount of **A** releases more HCl than **B**.
- E.** In solid **4**, there are no hydrogen bonds among the non-covalent interactions within the molecule.
- F.** The reason for the increase in viscosity of the chloroform solution of **4** with increasing concentration is the formation of one-dimensional supramolecular polymers by **4** molecules, mainly relying on hydrogen bonds rather than covalent reactions, with 5 intermolecular hydrogen bonds that can form between every two monomers.

Answer

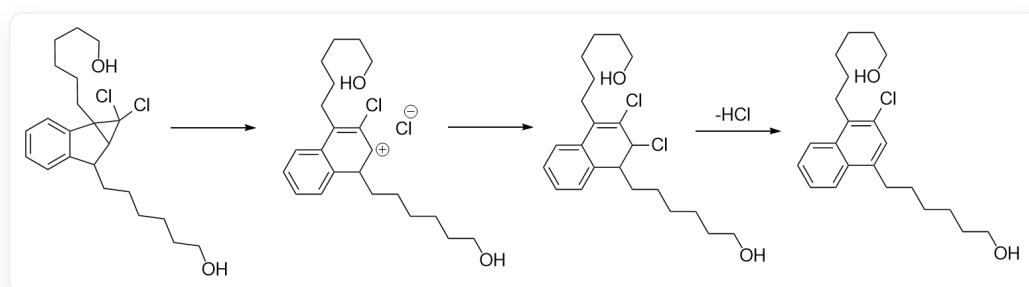
Correct Answer: **B**

Detailed Explanation

From the problem statement, heating **1** releases HCl and yields an oily liquid. Since **1** contains two chlorine atoms, it can produce at most two molecules of HCl. After releasing the first molecule of HCl, a stable naphthalene ring system can be formed in the product:



The mechanism of this process is that halogenated cyclopropanes undergo rearrangement under heating conditions to produce allylic halogenated products, which then rapidly aromatize to form a naphthalene ring, releasing one molecule of HCl under heating conditions.



机理为多步反应, ClC1(Cl)C2C(CCCCCCO)C3=CC=CC=C3C21CCCCC(O)>>ClC1=C(CCCCCCO)C2=CC=CC=C2(CCCCCCO)[CH+][1].[Cl-],
ClC1=C(CCCCCCO)C2=CC=CC=C2C(CCCCCCO)[CH+][1].[Cl-]>>ClC1=C(CCCCCCO)C2=CC=CC=C2(CCCCCCO)C1Cl,
ClC1=C(CCCCCCO)C2=CC=CC=C2C(CCCCCCO)C1Cl>>ClC1=C(CCCCCCO)C2=CC=CC=C2(CCCCCCO)=C1, 在加热条件下首先高张力的三元环开环, 离
 去一分子氯离子, 形成烯丙基正离子, 随后离子对反应, 产生烯丙基氯代物, 最后在芳构化的驱动下离去一分子HCl形成萘环

If the second molecule of HCl continues to be released, the degree of unsaturation of the system will be too high, and the activation energy provided at 50°C is insufficient to allow the Cl attached to the sp^2 carbon to depart. Therefore, the stoichiometric ratio of HCl produced by the complete reaction of **1** to the product molecule is 1:1.

CHECKPOINT

1 PTS

释放一分子HCl得到稳定的萘环，第二分子HCl的释放很困难，因此比例为1:1，选项A错误

The aromatic system contained in the product is a naphthalene ring, which has 10 π electrons.

CHECKPOINT

1 PTS

由于产物中有萘环，因此有10个 π 电子，选项B正确

Polymer **A** contains approximately 5% of **1** and 5% of **2**. The average functionality of the polymer is calculated as: $(0.9 \times 2 + 0.05 \times 4) \div 0.95 = 2.1$. Polymer **B** contains approximately 5% of **3**. The average functionality of the polymer is calculated as: $(0.95 \times 2 + 0.05 \times 4) \div 1 = 2.1$, both of which are greater than 2, and can form a covalently cross-linked network structure.

CHECKPOINT

1 PTS

聚合物**A**与**B**的平均官能度均大于2，均可形成网状结构，选项C错误

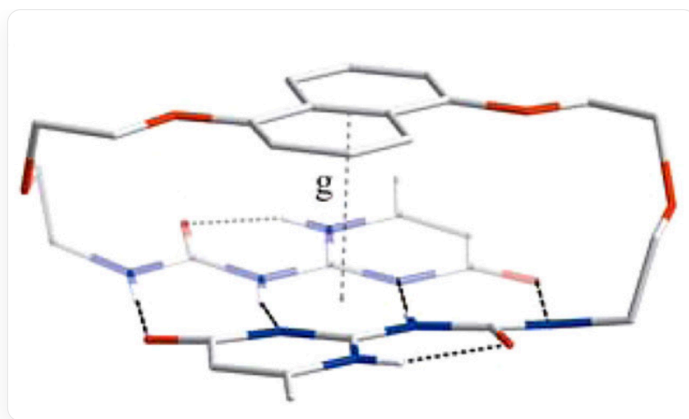
Since **1** cannot be inserted into the polymer network in a covalent manner, it can only be dispersed between the networks; while **3** can participate in the polymerization and be inserted into the polymer network, so B is more sensitive to pressure, and can produce more HCl under the same pressure.

CHECKPOINT

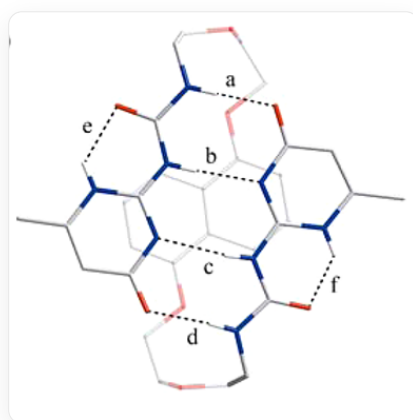
1 PTS

A中压力敏感基团分布在聚合物网络间而B中压力敏感基团参与聚合物网络形成，对压力响应更敏感，可释放更多HCl，选项D错误

4 forms a supramolecular polymer through intermolecular hydrogen bonds. In the solid-state monomer, intramolecular hydrogen bonds are formed instead of intermolecular hydrogen bonds to avoid polymerization.



单体两端的脲基与4-嘧啶酮形成分子内氢键，其中脲的两个氨基氢作为给体，4-嘧啶酮的氧和3号位的氮作为受体；4-嘧啶酮1号位氮的氢作为给体，脲的羰基氧作为受体，中心的萘环同时与两端的4-嘧啶酮存在 π - π 相互作用



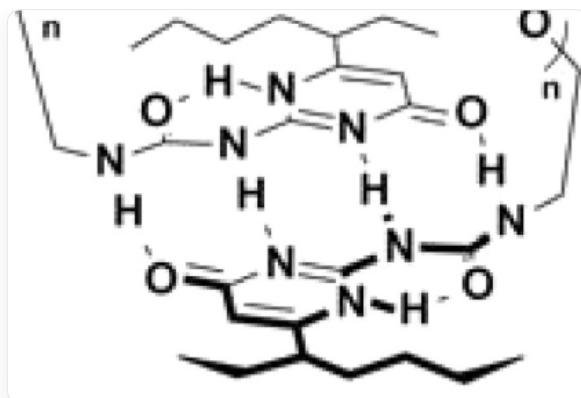
更清晰的展示出脲与4-嘧啶酮形成的氢键

CHECKPOINT

1 PTS

固态单体中同时存在氢键和 π - π 相互作用，选项E错误

After **4** is dissolved in chloroform, it can form a supramolecular polymer through hydrogen bonds. Under these conditions, the carbonyl group of urea will form an intramolecular six-membered ring hydrogen bond with the N-H of 4-pyrimidinone. The remaining two hydrogen bond donors and two hydrogen bond acceptors can form 4 hydrogen bonds between two monomers.



图中展示出组装形成超分子聚合物时，分子间形成4根氢键，给体为腺的两个氨基氢，受体为4-嘧啶酮的氧和3号位氮；分子内形成一根氢键，给体为4-嘧啶酮1号位氮上的氢，受体为腺的羰基氧

CHECKPOINT

1 PTS

两个单体间形成4根分子间氢键，选项F错误